	WVS			ANES		
	RCG	Cal.	OSR	RCG	Cal.	OSR
gpt-5.4-nano	1.272	1.150	0.994	1.370	1.232	1.173
gemini-flash-lite	1.218	1.126	0.840	1.353	1.178	0.838
Qwen3.5-9B	1.273	1.145	0.992	1.347	1.217	1.213
gemma-4-12B	1.199	1.137	0.793	1.332	1.194	0.942
gpt-oss-20b	1.269	1.140	1.013	1.346	1.202	1.144
Llama-3.3-70B	1.275	1.147	0.722	1.277	1.198	0.853
Marginals	1.158	–	0.989	1.250	–	1.078
Copula	1.041	–	0.946	1.091	–	0.976
LCA	1.105	–	0.959	1.143	–	0.950

Table 1: The respondent coverage gap (RCG) and the off-support ratio (OSR) for each generator on each dataset. Cal. is the RCG after marginal calibration. OSR is reported before calibration.

are coupled through a correlation matrix estimated with Ledoit–Wolf shrinkage (Ledoit and Wolf, 2004), as fitting uses fewer respondents than questions in about half of the cells. *LCA* draws $\boldsymbol{u}$ from a two-class latent class model (Goodman, 1974), that is, a mixture of two class-specific Marginals generators, fitted with StepMix (Morin et al., 2025). Because these baselines are fitted to the same respondents against which they are evaluated, they serve as references for what a generator can achieve with access to the cell's respondents, rather than as direct competitors to the LLMs (fitting details in Appendix I).

5 Results

5.1 Main Results

For each cell, we compute RCG and OSR for 50 random draws of 100 human respondents. We first average the scores over draws within each cell, then average the resulting cell-level scores across cells. We also confirm that RCG behaves as intended; in each cell c,[9] we replace the generated set $\mathcal{G}_c$ with a separate sample of 100 human respondents drawn from $\mathcal{H}_c$, disjoint from the 100 respondents used for evaluation, and compute RCG exactly as above. The average RCG is 1.001 (WVS) and 1.000 (ANES). The same replacement gives an average OSR of 1.001 (WVS) and 1.002 (ANES). Results under the hold-out fit setting are reported in Appendix I.

Table 1 shows the RCG and OSR for each generator and dataset (Appendix K shows RCG on WVS and ANES by the number of conditioning attributes L). All uncalibrated LLM RCG values in Table 1 exceed 1.0 and the values of all three baselines, including the simplest baseline, Marginals. For every LLM on both datasets, the 95% confidence interval for the RCG difference from Marginals excludes zero (Table 8 in Appendix J). Thus no LLM covers the human respondents as well as another human sample does (1.000–1.001), or even as well as Marginals, which ignores all dependence between questions. RCG shows no consistent ordering by model family, weight availability (proprietary vs. open-weight), or model size.

Calibrating marginal distributions. We next ask whether correcting per-question response distributions closes the respondent coverage gap. We use the empirical human marginals in each evaluation cell as oracle targets, applying quantile mapping (Gudmundsson et al., 2012; Cannon et al., 2015) to ordered responses and minimum-rewrite remapping to nominal responses (Appendix P). Calibration (Cal. in Table 1) reduces RCG for all six LLMs on both datasets. The calibrated RCG values range from 1.126 to 1.150 on WVS and from 1.178 to 1.232 on ANES. They are lower than those of Marginals but higher than the human reference and those of both Copula and LCA. All differences from the three baselines are statistically significant (Table 9 in Appendix J). Thus, this item-wise correction improves respondent coverage but does not restore human-level coverage.

5.2 Failure Modes

Using the uncalibrated OSR column of Table 1, we next examine how each LLM fails to cover the human respondents. The baselines (Marginals, Copula, and LCA) have OSR at or near one (0.946–1.078) on both datasets. RCG alone does not show how a model fails: on ANES, the six LLMs span 1.277–1.370 in RCG but 0.838–1.213 in OSR.

In both datasets, OSR splits the six LLMs into two groups: the upper group (gpt-oss-20b, gpt-5.4-nano, and Qwen3.5-9B) and the lower group (gemma-4-12B, gemini-flash-lite, and Llama-3.3-70B). The OSRs of the lower group are below one (0.722–0.840 on WVS and 0.838–0.942 on ANES) and the RCGs are above one. By the definitions in Section 3.2, this combination indicates contraction. In the upper group, the OSRs on ANES are above one (1.144–1.213), which is the off-support pattern. By contrast, on WVS, the OSRs are close to one (0.992–1.013).

[9] We use the cell only if it contains at least 200 human respondents. This leaves 203 of the 730 cells for WVS and 196 of the 603 cells for ANES.